Recall $\qquad x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau)\, y(t-\tau)\, d\tau$

$$\downarrow$$

$$\underbrace{(x*y)}(t) = (y*x)(t)$$

Associative $\qquad x(t) * (\underbrace{h_1(t) * h_2(t)}) = (x_1(t) * h_1(t)) * h_2(t)$

Proof :- $\qquad h_1(t) * h_2(t) = \underbrace{h_2(t) * h_1(t)}_{g(t)} = \int_{\lambda=-\infty}^{\infty} h_2(\lambda)\, h_1(t-\lambda)\, d\lambda.$

$$x(t) * g(t) = \int_{\tau=-\infty}^{\infty} x(\tau)\, \underbrace{g(t-\tau)}\, d\tau$$

$$= \int_{\tau=-\infty}^{\infty} x(\tau) \left[\int_{\lambda=-\infty}^{\infty} h_1(\lambda)\, h_1(t-\tau-\lambda)\, d\lambda \right] d\tau.$$

assumption for this Course $\longrightarrow$ order of integration can be changed

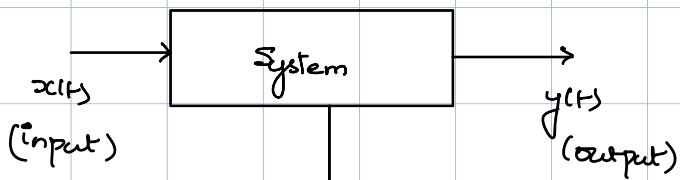
$$= \int_{\lambda=-\infty}^{\infty} h_2(\lambda) \underbrace{\left[\int_{\tau=-\infty}^{\infty} h_1(t-\lambda-\tau)\, x(\tau)\, d\tau \right]}_{(\underbrace{x*h_1}_{h(t-\lambda)})(t-\lambda)} d\lambda$$

$$= \int_{\lambda=-\infty}^{\infty} h_2(\lambda)\, h(t-\lambda)\, d\lambda = h_2(t) * (x(t) * h_1(t)) = (x_1(t) * h_1(t)) * h_2(t)$$

System :- Transforms a signal into another signal.

signal → input signal

another signal → output signal



Description → actual realization (RLC Ckt)

Description → mathematical model (✓)

$$y(t) = T(x(t))$$

indicator for how $x(t)$ transforms to $y(t)$

$y = T(x)$ or $y = f(x)$

Linear Vs Non-linear System :

A system $y = T(x)$ is linear if

a) $T(x_1 + x_2) = T(x_1) + T(x_2)$ (Additivity)

b) $T(\alpha x_1) = \alpha T(x_1)$ (homogenity) $\alpha \in \mathbb{C}$ → Fourier Series, Phasors

To check for linearity : $T(a x_1 + b x_2) = a T(x_1) + b T(x_2)$

If $T(x)$ satisfies additivity

a) $T(0) = 0$ $(\checkmark)$

b) $T(nx) = nT(x) \quad \forall n \in \mathbb{Z}$ ( )

c) $T(\frac{m}{n}x) = \frac{m}{n}T(x) \quad \forall m, n \in \mathbb{Z}, n \neq 0$ $\longrightarrow$ first Prove

d) $T(\alpha x) = \alpha T(x) \quad \forall \alpha \in \mathbb{R}$ if $T(x)$ is Cont at any Point x.

Proof :- if $n > 0 \Rightarrow T(x + x + \cdots x) = T(x) + \cdots + T(x) = nT(x)$

if $n = -1$ Need to show $T(-x) = -T(x)$ —

$$T(0) = T(x + \underbrace{(-x)}_{y}) = T(x) + T(-x) = 0$$

$$\Rightarrow T(-x) = -T(x)$$

Combine to show that $T(nx) = nT(x)$ $n \in \mathbb{Z}$

Examples:

① $y(t) = \frac{d}{dt}(x(t)) \longrightarrow$ Yes, linear.

② $y(t) = t\,x(t) \longrightarrow T(ax_1 + bx_2) = at\,x_1 + bt\,x_2 = t\,(ax_1 + bx_2)$ (Yes)

$$\underbrace{t\,(ax_1+bx_2)} \quad a\,\underbrace{tx_1} + b\,\underbrace{tx_2} \quad a\,y_1(t) + b\,y_2(t)$$

y is the o/p and x is the i/p.

Note:- No Constant term

③ $a_n \frac{d^n y}{dt^n} + \cdots + a_0 y = x(t) \longrightarrow$ Yes as long as $a_0, \cdots a_n$ are Constants.

(hint: next slide)

Prove before next lecture

④ $y(t) = x(t - t_0) \longrightarrow$ Yes

Prove $\underbrace{a_n \frac{d^n y}{dt^n} + \cdots + a_0 y}_{\sum_i a_i \frac{d^i y}{dt^i} = x(t)} = x(t)$ is linear when $a_0, \ldots a_n$ are Constants

Proof:

Let $y_1(t)$ Satisfy DE when i/p is $x_1(t)$ $\quad x_1 \to \boxed{\text{System}} \to y_1$

Let $y_2(t)$ " " " " " $x_2(t)$ $\quad x_2 \to \boxed{\text{System}} \to y_2$

Easy to Prove that

if we define $y = \alpha y_1 + \beta y_2$ then y is a solution to

$x = \alpha x_1 + \beta x_2$

$\downarrow$

$$a_n \frac{d^n(\alpha y_1 + \beta y_2)}{dt^n} + \cdots + a_0(\alpha y_1 + \beta y_2)$$

$$= \alpha \underbrace{\left[a_n \frac{d^n y_1}{dt^n} + \cdots + a_0 y_1\right]}_{x_1(t)} + \beta \underbrace{\left[a_n \frac{d^n y_2}{dt^n} + \cdots + a_0 y_2\right]}_{x_2(t)}$$

$$= \alpha x_1(t) + \beta x_2(t)$$

Complete Proof ??